University of Minnesota custom Azure course

Lab 3: Immutable Backup Vault with Multi-User Authorization

Participant lab guide

Session 3: Data Protection in Both Directions

Session: Day 2, Session 3, Block 4 (75 minutes: 55 lab, 20 debrief)

Format: Individual, hands-on, in your two lab resource groups

Outcome: A Backup vault that refuses its own administrator, a vaulted blob backup, and a restore that proves it

All data in this lab is synthetic. No real University of Minnesota data is used or required.

Contents

1. Lab overview 3
1.1 Learning objectives 3
1.2 Timing 3
1.3 Prerequisites 3
2. Part 0: sign in and set your variables (4 minutes) 4
3. Part 1: Resource Guard in the guard resource group (6 minutes) 4
4. Part 2: Backup vault with immutability, soft delete, and MUA (10 minutes) 5
5. Part 3: the storage account to protect (8 minutes) 6
6. Part 4: vaulted blob backup (12 minutes, then it runs in the background) 7
7. Part 5: protected operations, refused (8 minutes) 8
8. Part 6: restore to an alternate storage account (7 minutes) 9
9. Debrief questions 11
10. Worksheet 11
11. Reference 12

1. Lab overview

In Session 3 you watched five attack paths fail against an immutable, soft-delete-protected, multi-user-authorized vault. In this lab you build the Backup vault version of that defense yourself: a Resource Guard owned by a separate resource group, a Backup vault with immutability enabled and linked to the guard, vaulted backup of a storage account's blob containers, a protected operation that the vault refuses until the guard authorizes it, and a restore to an alternate storage account that proves the backup is usable.

The vault stays **Unlocked** so the lab can be cleaned up. Locked immutability is irreversible, which is exactly why it is the production setting and not the lab setting.

1.1 Learning objectives

- Deploy a Resource Guard in a separate resource group and explain why separation is the point.
- Create a Backup vault with immutability and soft delete, with a system-assigned identity, and link it to the Resource Guard (multi-user authorization).
- Grant the vault identity the Storage Account Backup Contributor role and configure vaulted backup for selected containers.
- Attempt a protected operation and interpret the refusal.
- Restore a container to an alternate storage account from a vaulted recovery point.
- Read the backup jobs and the alert configuration that an operator would use day to day.

1.2 Timing

Part	Activity	Minutes
0	Sign in and set your variables	4
1	Resource Guard in the guard resource group	6
2	Backup vault with immutability and soft delete, linked to the guard	10
3	Storage account to protect, seeded with two containers	8
4	Vaulted blob backup: role, configuration, backup instance, on-demand backup (long-running; started early)	12
5	Protected operations: watch the vault refuse you	8
6	Restore to an alternate storage account and read the jobs	7
	Debrief (whole room)	20

1.3 Prerequisites

- Two lab resource groups assigned by the instructor: `rg-lab-NN` (your vault group) and `rg-lab-NN-guard` (your guard group). In production the guard group belongs to Security; in the lab you own both so you can see both sides.
- Azure CLI 2.70 or later with the `dataprotection` extension (`az extension add --name dataprotection`), or Azure Cloud Shell.
- The lab files folder `lab-solutions/s03-immutable-backup-vault` for the optional as-code path.

Conventions

Replace `NN` with your lab number and `<suffix>` with your initials plus two digits in lowercase. Every part ends with a **Checkpoint**.

Vaulted blob backup uses a two-step initialization in the CLI: a backup configuration that names the containers, then a backup instance that references it. The guide shows both.

The first vaulted backup can take 20 to 40 minutes. Part 4 starts it; Part 6 uses its recovery point. If it has not finished by Part 6, run the restore after the debrief; the commands are the same.

2. Part 0: sign in and set your variables (4 minutes)

```
az login
az extension add --name dataprotection --upgrade
export RG=rg-lab-NN
export RGG=rg-lab-NN-guard
export LOCATION=$(az group show -n $RG --query location -o tsv)
export SUFFIX=<suffix>
export VAULT=bvault-lab03-$SUFFIX
export SA=stlab03$SUFFIX
export ME=$(az ad signed-in-user show --query id -o tsv)
echo "Vault: $VAULT  Account: $SA  Region: $LOCATION"
```

Checkpoint 0

Both resource groups show with `az group show -n $RG` and `az group show -n $RGG`, and the extension list includes `dataprotection`.

3. Part 1: Resource Guard in the guard resource group (6 minutes)

A Resource Guard protects a set of critical vault operations: the vault administrator cannot perform them unless they also hold a role (Reader) on the guard. The guard therefore lives where the vault administrator has no rights. The lab's second resource group stands in for a separate subscription or tenant.

Portal path

1. Portal > search **Resource Guards** > **Create**. Resource group `rg-lab-NN-guard`, name `guard-lab03-<suffix>`, region your lab region.
2. Protected operations tab: leave every operation selected (none excluded). Review and create.

CLI path

```
az dataprotection resource-guard create -g $RGG --resource-guard-name guard-lab03-$SUFFIX -
l $LOCATION \
  --tags Owner=security-team Project=UMN-Azure-Course -o table
export GUARD_ID=$(az dataprotection resource-guard show -g $RGG --resource-guard-name
guard-lab03-$SUFFIX --query id -o tsv)
az dataprotection resource-guard list-protected-operations -g $RGG --resource-guard-name
guard-lab03-$SUFFIX \
  --resource-type Microsoft.DataProtection/backupVaults -o table
```

> **Checkpoint 1**
> `$GUARD_ID` is set and the protected-operations list for Backup vaults shows the critical operations (deleting backup instances, disabling soft delete, disabling immutability, reducing retention, and related operations). Write two of them on your worksheet.

4. Part 2: Backup vault with immutability, soft delete, and MUA (10 minutes)

4.1 Create the vault

Vault store redundancy is set at creation and cannot be changed after the first item is protected. The lab uses locally redundant storage to keep cost low; the design exercise discusses when geo-redundant is right.

```
az dataprotection backup-vault create -g $RG --vault-name $VAULT -l $LOCATION \
  --type SystemAssigned \
  --storage-setting "[{type:'LocallyRedundant',datastore-type:'VaultStore'}]" \
  --immutability-state Unlocked \
  --soft-delete-state On --retention-duration-in-days 14 \
  --azure-monitor-alerts-for-job-failures Enabled \
  --tags Project=UMN-Azure-Course Owner=lab-NN -o table
export VAULT_PRINCIPAL=$(az dataprotection backup-vault show -g $RG --vault-name $VAULT --
query identity.principalId -o tsv)
```

Portal equivalent: Backup center > Vaults > **+ Vault** > Backup vault; on the Vault properties tab choose Locally-redundant; after creation open **Properties** and set Immutability to **Enabled** (unlocked) and Soft delete to **On**, 14 days.

4.2 Link the vault to the Resource Guard

```
az dataprotection backup-vault resource-guard-mapping create -g $RG --vault-name $VAULT \
  -n DppResourceGuardProxy --resource-guard-id $GUARD_ID -o table
```

```
az dataprotection backup-vault resource-guard-mapping show -g $RG --vault-name $VAULT -n
DppResourceGuardProxy \
  --query "properties.resourceGuardResourceId" -o tsv
```

Portal equivalent: vault > **Properties** > Multi-user authorization > **Update** > select the Resource Guard. The link succeeds only because you hold Reader (through Contributor) on the guard group; a vault administrator without that role cannot create it, which is the first protected operation you benefit from.

Checkpoint 2

`az dataprotection backup-vault show -g $RG --vault-name $VAULT --query "{immutability:properties.securitySettings.immutabilitySettings.state, softDelete:properties.securitySettings.softDeleteSettings.state, days:properties.securitySettings.softDeleteSettings.retentionDurationInDays}" -o table` shows `Unlocked`, `On`, `14`, and the mapping show prints the guard's resource ID.

5. Part 3: the storage account to protect (8 minutes)

Vaulted blob backup needs a storage account with at least one container. The account below follows the security baseline and enables versioning and change feed, which vaulted backup does not require but the operational tier does; enabling both lets you compare the tiers in the debrief.

```
az storage account create -n $SA -g $RG -l $LOCATION --kind StorageV2 --sku Standard_ZRS \
  --https-only true --min-tls-version TLS1_2 --allow-blob-public-access false --allow-
shared-key-access false \
  --tags ResearchUnit=GSC DataClass=GenomicSequences RetentionClass=Project Owner=lab-NN
az storage account blob-service-properties update -n $SA -g $RG --enable-versioning true --
enable-change-feed true \
  --enable-delete-retention true --delete-retention-days 7 --enable-container-delete-
retention true --container-delete-retention-days 7
export SA_ID=$(az storage account show -n $SA -g $RG --query id -o tsv)

az role assignment create --role "Storage Blob Data Contributor" --assignee-object-id $ME -
-assignee-principal-type User --scope $SA_ID -o none
sleep 60
az storage container create --account-name $SA -n research-active --auth-mode login -o none
az storage container create --account-name $SA -n research-reference --auth-mode login -o
none
mkdir -p ~/lab03/a ~/lab03/r
for i in 1 2 3 4 5; do head -c 524288 /dev/urandom > ~/lab03/a/sample_active_$i.bin; head -
c 524288 /dev/urandom > ~/lab03/r/sample_reference_$i.bin; done
az storage blob upload-batch --account-name $SA --auth-mode login -d research-active -s
~/lab03/a -o none
az storage blob upload-batch --account-name $SA --auth-mode login -d research-reference -s
~/lab03/r -o none
```

Checkpoint 3

> `az storage blob list --account-name $SA -c research-active --auth-mode login --query "length(@)"` returns 5, and the same for `research-reference`.

6. Part 4: vaulted blob backup (12 minutes, then it runs in the background)

6.1 Grant the vault identity its role

The vault's managed identity needs **Storage Account Backup Contributor** on the account. Microsoft Learn notes the assignment can take up to 30 minutes to take effect; in dry runs it was usually under five.

```
az role assignment create --role "Storage Account Backup Contributor" \
  --assignee-object-id $VAULT_PRINCIPAL --assignee-principal-type ServicePrincipal --scope
$SA_ID -o table
```

6.2 Create the vaulted backup policy

Daily, retained 30 days. The policy template from the CLI is the operational-tier template; the vaulted policy below is written explicitly so you can read every field: a scheduled trigger, a `VaultStore` data store, and a retention rule with an absolute delete option.

```
cat > blob-policy.json <<'EOF'
{
  "objectType": "BackupPolicy",
  "datasourceTypes": [ "Microsoft.Storage/storageAccounts/blobServices" ],
  "policyRules": [
    {
      "name": "BackupDaily", "objectType": "AzureBackupRule",
      "backupParameters": { "objectType": "AzureBackupParams", "backupType": "Discrete" },
      "dataStore": { "dataStoreType": "VaultStore", "objectType": "DataStoreInfoBase" },
      "trigger": {
        "objectType": "ScheduleBasedTriggerContext",
        "schedule": { "repeatingTimeIntervals": [ "R/2026-10-20T06:00:00+00:00/P1D" ],
"timeZone": "UTC" },
        "taggingCriteria": [ { "tagInfo": { "tagName": "Default" }, "taggingPriority": 99,
"isDefault": true } ]
      }
    },
    {
      "name": "Default", "objectType": "AzureRetentionRule", "isDefault": true,
      "lifecycles": [ {
        "deleteAfter": { "objectType": "AbsoluteDeleteOption", "duration": "P30D" },
        "sourceDataStore": { "dataStoreType": "VaultStore", "objectType":
"DataStoreInfoBase" },
        "targetDataStoreCopySettings": []
      } ]
    }
  ]
}
EOF
```

```
az dataprotection backup-policy create -g $RG --vault-name $VAULT -n blob-vaulted-daily --
policy blob-policy.json -o table
export POLICY_ID=$(az dataprotection backup-policy show -g $RG --vault-name $VAULT -n blob-
vaulted-daily --query id -o tsv)
```

6.3 Configure the backup instance (two steps) and start a backup

```
# Step 1: the backup configuration names the containers for the vault store
az dataprotection backup-instance initialize-backupconfig --datasource-type AzureBlob \
  --storage-account-rg $RG --storage-account-name $SA \
  --container-list research-active research-reference > backup-config.json

# Step 2: the backup instance references the policy, the account, and that configuration
az dataprotection backup-instance initialize --datasource-type AzureBlob --datasource-
location $LOCATION \
  --policy-id $POLICY_ID --datasource-id $SA_ID --backup-configuration backup-config.json >
backup-instance.json

# Validate (fails until the role assignment has propagated; retry every few minutes)
az dataprotection backup-instance validate-for-backup -g $RG --vault-name $VAULT --backup-
instance backup-instance.json

# Create, then find the generated instance name and start an on-demand backup
az dataprotection backup-instance create -g $RG --vault-name $VAULT --backup-instance
backup-instance.json -o table
export BI=$(az dataprotection backup-instance list -g $RG --vault-name $VAULT --query
"[0].name" -o tsv)
az dataprotection backup-instance adhoc-backup -g $RG --vault-name $VAULT -n $BI --rule-
name BackupDaily --no-wait
date -u
```

Portal equivalent: vault > **Backup** > Datasource type Azure Blobs (Azure Storage) > select the account > **Vaulted backup** > choose containers > policy `blob-vaulted-daily` > Backup. Then Backup instances > the account > **Backup now**.

> **Checkpoint 4**
> `az dataprotection job list -g $RG --vault-name $VAULT --query "[].{op:properties.operationCategory,status:properties.status,start:properties.startTime}" -o table` shows a Backup job `InProgress` (or `Completed` if you are quick). Write the start time on your worksheet and continue; Part 6 needs this job to finish.

7. Part 5: protected operations, refused (8 minutes)

You are the vault administrator and you hold Reader on the guard through your Contributor role on the guard group. To experience the refusal, you must temporarily become an administrator who does **not**: remove your own access to the guard resource group, try the protected operation, then put the access back. (In production the vault admin never had it.)

```
export RGG_ID=$(az group show -n $RGG --query id -o tsv)
# List what you hold on the guard group, then remove it
az role assignment list --assignee $ME --scope $RGG_ID -o table
az role assignment delete --assignee $ME --role Contributor --scope $RGG_ID
az role assignment delete --assignee $ME --role "User Access Administrator" --scope $RGG_ID
2>/dev/null || true
sleep 90

# Protected operation 1: disable immutability
az dataprotection backup-vault update -g $RG --vault-name $VAULT --immutability-state
Disabled
# Protected operation 2: turn soft delete off
az dataprotection backup-vault update -g $RG --vault-name $VAULT --soft-delete-state Off
# Protected operation 3: delete the backup instance (and its recovery points)
az dataprotection backup-instance delete -g $RG --vault-name $VAULT -n $BI --yes
```

Expected: each request is rejected with an authorization error that names the Resource Guard (the caller lacks the required permission on the guard). The vault, the instance, and the settings are unchanged.

Restore your guard access before continuing

Ask the instructor to re-grant Contributor on `rg-lab-NN-guard` (the instructor holds Owner on the subscription), or if you were given User Access Administrator at the subscription scope for the course, re-add it yourself: `az role assignment create --role Contributor --assignee-object-id $ME --assignee-principal-type User --scope $RGG_ID`.

Checkpoint 5

`az dataprotection backup-vault show -g $RG --vault-name $VAULT --query "properties.securitySettings"` still shows immutability `Unlocked` and soft delete `On`, and `az dataprotection backup-instance list -g $RG --vault-name $VAULT --query "length(@)"` returns 1. Write the first line of one refusal message on your worksheet.

8. Part 6: restore to an alternate storage account (7 minutes)

Vaulted recovery points are restored to a storage account other than the source, which keeps the restore from overwriting anything. The vault identity needs Storage Account Backup Contributor on the target too.

```
# Wait for the backup to complete
az dataprotection job list -g $RG --vault-name $VAULT --query
"[?properties.operationCategory=='Backup'].{status:properties.status,end:properties.endTime}"
-o table

# Alternate account
export ALT=stlab03alt$SUFFIX
```

```
export ALT_ID=$(az storage account create -n $ALT -g $RG -l $LOCATION --kind StorageV2 --sku
Standard_LRS \
  --https-only true --min-tls-version TLS1_2 --allow-blob-public-access false --query id -o
tsv)
az role assignment create --role "Storage Account Backup Contributor" --assignee-object-id
$VAULT_PRINCIPAL \
  --assignee-principal-type ServicePrincipal --scope $ALT_ID -o none
sleep 60

# Newest recovery point, then restore one container to the alternate account
export RP=$(az dataprotection recovery-point list -g $RG --vault-name $VAULT --backup-
instance-name $BI --query "[0].name" -o tsv)
az dataprotection backup-instance restore initialize-for-item-recovery --datasource-type
AzureBlob \
  --restore-location $LOCATION --source-datastore VaultStore --recovery-point-id $RP \
  --target-resource-id $ALT_ID --container-list research-active > restore.json
az dataprotection backup-instance validate-for-restore -g $RG --vault-name $VAULT -n $BI --
restore-request-object restore.json
az dataprotection backup-instance restore trigger -g $RG --vault-name $VAULT -n $BI --
restore-request-object restore.json
az dataprotection job list -g $RG --vault-name $VAULT --query
"[].{op:properties.operationCategory,status:properties.status}" -o table
```

Portal equivalent: vault > Backup instances > the account > **Restore** > select the recovery point > restore target **Alternate** storage account > select containers > Restore. Then verify with `az storage blob list --account-name $ALT -c research-active --auth-mode login --query "length(@)"` once the job completes (you need Storage Blob Data Reader on the alternate account to list; grant it to yourself).

> **Checkpoint 6**
> A Restore job appears in the job list. When it completes, the alternate account holds `research-active` with five blobs. If the backup is still running at the end of the lab, finish this part after the debrief; the recovery point will be there.

8.1 Clean up

Immutability (Unlocked) and soft delete both resist deletion by design. The order: delete the backup instance (a protected operation, so your guard access must be back), disable immutability, then delete the vault after the soft delete window, or leave the vault for the instructor's cleanup pass. The storage accounts and the Resource Guard can go now.

```
az dataprotection backup-instance delete -g $RG --vault-name $VAULT -n $BI --yes
az dataprotection backup-vault resource-guard-mapping delete -g $RG --vault-name $VAULT -n
DppResourceGuardProxy --yes
az dataprotection backup-vault update -g $RG --vault-name $VAULT --immutability-state
Disabled
az storage account delete -n $ALT -g $RG --yes
# The protected account carries a Backup-owned delete lock while protected; remove it after
the instance is deleted:
az lock list --resource-group $RG --resource-type Microsoft.Storage/storageAccounts --
resource $SA -o table
```

9. Debrief questions

1. Which control would have stopped the attack you worry about most in the university's environment? Name the attack and the control.
2. Who in the university should own the Resource Guard, and where should it live (subscription, tenant)? What has to be true about that team's access to the vaults?
3. What would you tell your vice president "immutable" means now? One sentence.
4. Vaulted backup kept a copy outside the source account; the operational tier would not have. When is operational backup still the right choice, and for which of the university's data?
5. The first backup took a while. What does that imply for how you would sequence protection across several petabytes, and for the restore test cadence?

Notes for the debrief

10. Worksheet

Item	Your value or observation
Resource Guard name and two protected operations	
Backup vault name, redundancy, immutability state	

Item	Your value or observation
Time the on-demand backup started (UTC)	
Time it completed	
First line of one refusal message in Part 5	
Recovery point name used for the restore	
Blob count in the alternate account after restore	

11. Reference

- Immutable vault for Azure Backup (Microsoft Learn)
- Multi-user authorization using Resource Guard (Microsoft Learn)
- Soft delete and enhanced soft delete for Azure Backup (Microsoft Learn)
- Configure vaulted backup for Azure Blobs using Azure CLI; Restore Azure Blobs using Azure CLI (Microsoft Learn)
- Azure CLI reference: `az dataprotection backup-vault`, `backup-policy`, `backup-instance`, `resource-guard`, `recovery-point`